Shock and Contagion

Self-Exciting Point Processes for Surges in Antisemitic Incidents

IN ONE LINE

When antisemitic incidents spike after a major event, our tools estimate how much of the spike is a direct reaction and how much is incidents setting off more incidents, and say clearly when the data cannot tell the two apart.

THE PROBLEM

Antisemitic incidents arrive in surges, often after salient events: the UK's Community Security Trust recorded an average of 5 incidents a day in 2023 before 7 October, and 31 a day from 7 October to the end of the year. A surge raises three questions: how much is a direct reaction to the event, how much is incidents prompting further incidents through imitation or emboldenment, and does online hostility spill over into offline incidents? The answers call for different responses, from vigilance after similar events to breaking chains early or watching online activity for early warning. Handled carelessly, the same data can seem to show "contagion" that is not there, which matters to any community that plans its security around such patterns.

WHAT WE BUILT

- **A "ripple" model of incidents** (a Hawkes process). Each incident is either a response to background conditions or to an outside event, or is triggered by an earlier incident, possibly across regions or between online and offline channels.
- **A surge breakdown.** The expected number of incidents after an event is split into what would have happened anyway, the direct reaction, and the knock-on cascade, each with a margin of uncertainty.
- **Three traps, proven and named.** Within a single stream of incidents, cascades and clusters of unrecorded outside shocks can look identical; a model that misses events or is too rigid "finds" contagion that is not there; and under-reporting hides real contagion, which we correct using Paper 1's estimate of how many incidents go unrecorded.
- **An early-warning rule** that makes the trade-off between false alarms and missed surges an explicit choice, agreed with the people who bear the costs, plus an adjustment for reporting delays.

WHAT OUR SIMULATIONS SHOW

All results come from computer-simulated incident data in which we know the true answer.

- **Phantom contagion is a real risk.** In data with no contagion at all, a model with a fixed background rate attributed up to 63% of incidents to earlier incidents. Smoothing the background did not remove this when the hidden shocks were short and sharp; supplying the dates of the outside events largely did, where tested.
- **Correcting for under-reporting mostly works, within limits.** It restored accurate uncertainty ranges except when contagion was weak and few incidents were recorded. If the recording rate itself is uncertain by 30–50%, that sets a floor on precision that more data cannot lower.
- **No model sees a new event coming.** A model-based alert rule beat two simple baselines, but on data generated by that same model, a favourable test. In 62% of simulated surges a new event occurred after the forecast was made; the rule caught 28% of those surges, against 89% of the others.

WHAT THIS IS NOT

STATUS **A methods paper, with no real-world findings.** No real contagion has been measured yet: nothing here shows that antisemitic incidents are, or are not, contagious. The figures under "The problem" only motivate the questions; every result comes from simulated data. The paper proposes how to measure these patterns responsibly.

WHAT WOULD COME NEXT

An analysis of New York City police records of anti-Jewish hate crimes from 2019 on, borough by borough, with a list of salient events fixed by coders who never see the incident data, and a plan registered publicly before any outcome data are examined. Contagion will be claimed only if it passes several pre-set tests, online–offline links will be reported as predictive, never causal, and any alerts would go only to vetted community-security partners, never published or mapped below borough level.

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